

The Hidden Palette: Uncovering Fashion Shopping Trends

1. Project Overview

Goal/Research Question: What attribute combinations frequently occur together across fashion products? Can we uncover fashion item clusters using categorical features?

Our main goal is to uncover patterns in fashion data that reveal which items naturally belong together and how trends evolve across seasons and years. We aim to analyze attributes such as color, season, and category to identify meaningful clusters. For example, whether bright tees dominate summer collections or dark outerwear appears more frequently in winter. This aligns directly with our approved proposal, which focuses on discovering co-occurring fashion attributes and understanding how product groupings form over time.

Motivation and Significance

Fashion may seem subjective, but the industry relies heavily on predicting what consumers will want next. However, without data mining, these inferences are often guided by intuition, limited historical insight, or incomplete information. By systematically analyzing fashion data, we can identify consistent relationships (e.g., colors tied to specific seasons, styles that rise together, categories that frequently co-occur). Understanding these patterns moves trend forecasting from guesswork to evidence-based analysis.

By transforming basic fashion related attributes into interpretable patterns, our work supports the broader objective of analyzing how trends shift and how these insights can inform real-world applications like inventory planning, palette selection, and improved recommender systems.

A key significance of our project is that we are applying symbolic market basket and categorical clustering analysis to a dataset that is typically used for deep learning image-based modeling. This makes our analysis valuable as a supplementary perspective to the deep learning workflows usually built on top of this dataset.

Addressing Challenges

In our initial proposal, we identified two key limitations: (1) the difficulty of finding meaningful patterns given uneven year distribution, and (2) the presence of redundant text fields such as *productDisplayName*. During the project, we adjusted our approach to resolve both issues effectively.

1. Uneven Year Distribution: The dataset had a heavy concentration of products in a few specific years, which made year-to-year comparisons unreliable. To avoid drawing misleading conclusions from this imbalance, we decided not to use yearly trend analysis at all.

Instead, we shifted to examining seasonal patterns, which are more consistent and better represented throughout the data. This allowed us to focus on how certain item types or colors appear across seasons (e.g., summer vs. winter) without being affected by the uneven spread of

years. This adjustment ensured that our findings reflected genuine fashion patterns rather than artifacts of the dataset distribution.

2. Long Text Fields (e.g., *productDisplayName*): Another limitation was the potential noise in text-heavy fields like *productDisplayName*, which included brand names. While we could have extracted some interesting insights from extracting the brand name from the column, we decided against using the field in its entirety (See Section 3: Dataset and Representation - *Data Description and Integrity*).

We also chose not to use *masterCategory* or *subCategory*, because we found that *articleType* alone captured most of the useful, specific information we needed, such as whether an item was a tee, a dress, a coat, etc. Using *articleType* as our primary product descriptor kept our analysis simple, cleaner and more interpretable.

2. Background / Related Work

Relevant Prior Studies

One closely related study is “*Association Rules for Product Sales Data Analysis Using the Apriori Algorithm*” by Pamungkas and Handrianto (2020)[1]. The authors apply the Apriori algorithm to retail transaction data to uncover products that are frequently purchased together. Their analysis identifies dominant item pairs, such as shirts and pants, with high support and confidence values. By using association rule mining, the study demonstrates how retailers can gain actionable insights into customer purchasing behavior, optimize product placement, and design more effective sales strategies. Although their dataset comes from a traditional retail environment and focuses primarily on transactional co-purchases, the underlying goal of extracting meaningful relationships from catalogued items directly parallels the motivations of our own work.

Most research on our chosen dataset from Kaggle focuses on computer vision tasks such as classification, feature extraction, or model benchmarking using deep learning architectures (See *Fashion Product Classification through Deep Learning and Computer Vision*” by Donati et al. (2019)[2]). These studies treat the dataset primarily as an image-based challenge and rarely explore its tabular or categorical metadata.

Finally, while this is not a prior study, this project’s topic of **Fashion** was inspired by a team member’s - Robbie Collis- project from a previous semester: Threadlines[3]. ThreadLines is a data-driven exploration of men’s fashion brands using Reddit discussions, natural language processing, clustering, and graph analysis. By mining adjectives and style cues from posts on r/malefashionadvice, the project builds semantic and visual relationships between brands. It applies NLP techniques (spaCy) to extract descriptive terms, cosine similarity and K-Means clustering to group brands, and NetworkX to visualize a network of stylistic connections. The results are further summarized with word clouds and charts, highlighting key attributes associated with each brand. This project also relates to an application of data mining - Networks.

Connection to Current Work

Significant limitations of our cited similar work[1] includes their focus on only pairwise itemsets, and using Tanagra (a GUI-based tool), which possibly limits scalability compared to more programmatic or industrial-grade tools. Our work extends the approach[1] by applying association rule mining to symbolic fashion attributes - season, color, gender, and article type - rather than transactional purchase data. While their study identifies which products are frequently bought together, our analysis captures latent co-occurrence patterns across product characteristics, effectively creating “market baskets” of fashion attributes. Our results also address the limitations of the paper, since our work will be highly reproducible and focus on more than just two-item itemsets.

Our dataset has the limitation of being a synthetically produced dataset for Deep Learning purposes[2], but that does not work against our favor. Our work provides a supplementary analytical layer that can complement deep learning applications that use this dataset. For example, item-to-season relationships derived from Apriori could inform downstream feature engineering.

Finally, unlike Threadlines, our project does not depend on scraped data from a popular social media site like Reddit, where each individual user’s consent to us, accessing their data, is ambiguous at best - and is also advised against by the instructional team. Rather, our project contributes further to finding valuable insights about fashion - particularly Indian fashion - without having the need to utilize web scrapping.

3. Dataset & Representation

Data Description and Integrity

Our project uses the Fashion Product Images (Small) dataset[4] created by Param Aggarwal and hosted on Kaggle. The dataset was originally compiled from an Indian e-commerce platform and is frequently used to learn computer vision for clothing classification, attribute prediction, and deep learning model benchmarking[5].

The dataset contains approximately 44,000 fashion product entries, each consisting of a product image and a set of structured metadata fields. These fields describe various product attributes, including:

- *id*: Unique product identifier
- *gender*: Intended audience category (e.g., *Men*, *Women*, *Unisex*)
- *masterCategory*: High-level product grouping (e.g., *Apparel*, *Accessories*)
- *subCategory*: More specific category within the master category
- *articleType*: Item-level label describing the product type (e.g., *Tshirt*, *Dress*, *Jeans*)
- *baseColour*: Primary color of the product
- *season*: Retail season (e.g., *Summer*, *Fall*, *Winter*)
- *year*: Year of product release
- *usage*: Occasional or contextual category (e.g., *Sports*, *Casual*)
- *productDisplayName*: Text description used by the retailer

Images of the items are provided in a uniform format in an accompanying directory and are linked to metadata via the product ID. The dataset is stored as a single CSV file containing all metadata attributes and a folder of associated JPEG images.

The dataset is publicly available through Kaggle for non-commercial research and educational use. All images originate from a commercial catalog but are republished under a permissive license for academic analysis[6]. Our chosen dataset also addresses ethical and accessibility considerations as follows:

- The dataset contains only product images reducing typical privacy risks. Therefore, there are no human subjects personally or identifiable information.
- The dataset reflects the structure and biases of an Indian commercial fashion marketplace; therefore, class imbalance (e.g., more women’s products, popular categories overrepresented) may influence downstream results.
- All data is accessible in standard formats (CSV and JPEG) and can be processed using common open-source tools, making the dataset broadly usable in academic contexts.

Data Processing and Preparation

To ensure consistency across categorical attributes, we applied a series of very standard data-cleaning operations. First, we converted all string-based fields to lowercase and removed leading or trailing whitespace, preventing duplicate categories arising from formatting inconsistencies (e.g., “Fall” vs. “fall”). We then addressed missing values by replacing null entries with the placeholder “unknown”, allowing the dataset to remain fully usable during association rule mining. This approach avoids discarding observations, preserves dataset size, and ensures that missingness is treated as an interpretable category rather than causing computational issues during pattern extraction.

Unlike common DM/ML applications using this dataset[7], we will not be using the provided JPEG images in our analysis. Likewise, we intentionally excluded several metadata attributes from our analysis to improve the quality and interpretability of the resulting itemsets. *masterCategory* and *subCategory* were removed because they largely duplicate information already captured by *articleType*, offering little additional value. The *usage* attribute was excluded due to its highly imbalanced distribution, which would cause a single dominant category to overwhelm the frequent itemsets. We also omitted the *year*, as its high granularity and uneven distribution would fragment the data into many low-support combinations, reducing both the stability and meaningfulness of discovered patterns.

We also decided to exclude the brand name, which could have been extracted with *productDisplayName* to reduce obvious representations (*heel brands - women - heels, sport brands - sport wear*, etc.). Since article type, gender, color, and season already capture the core symbolic structure of the products, adding brands would not improve the representativeness of the baskets or enhance our ability to identify seasonal/style patterns to a significant extent.

To prepare the dataset for association rule mining, we standardized the selected attributes by converting all values to string type, ensuring uniform processing across categories. Then, each product was transformed into a set of attribute values, following the transactional format required by the Apriori algorithm. In this representation, each transaction corresponds to a single product, and its attribute set functions as the items used to identify frequent co-occurrence patterns. An example attribute set is as

follows: (44424, {'fall', 'men', 'navy blue', 'shirts'}). It is consistently ordered by season, gender, color, and article type.

To prepare the attribute-based transactions for K-Means clustering, we converted the categorical attribute sets into a binary feature matrix using the *TransactionEncoder*. This encoder creates one Boolean column for each unique attribute value in the dataset (e.g., “winter”, “red”, “men”, “t-shirt”), and marks each product with *True* or *False* depending on whether that attribute is present. Because the resulting matrix is sparse and high-dimensional, we employed Principal Component Analysis (PCA) to project the data onto a lower-dimensional space that captures the largest variance in these binary indicators. This implementation renders our features into a better fit for K-Means clustering, despite K-Means’ more common use for numerical clustering.

4. Analytical Techniques & Implementation

Appropriateness of Methods: We use three analytical techniques in our project.

1. Association Rule Mining (Apriori): Apriori is ideal for identifying frequent co-occurring attributes in categorical data. By calculating support, confidence, and lift, it reveals meaningful relationships between fashion attributes, allowing us to discover which combinations of features naturally occur together. This method is particularly well-suited for our goal of uncovering interpretable insights about what informs fashion trends.
2. Dimensionality Reduction (Multiple Correspondence Analysis, MCA): Dimensionality reduction techniques are used to project the high-dimensional binary attribute space into a lower-dimensional representation. This preserves the similarity structure among products so that items sharing attributes appear close together in visualizations. Since our dataset consists of many categorical attributes encoded as binary features, MCA is ideal for dimensionality reduction. Unlike PCA, which assumes continuous variables, MCA is specifically designed to capture and explain the variance in categorical attributes.
3. Clustering (K-Means): K-Means is an effective method for grouping products based on shared attribute presence. Using the binary-encoded attribute matrix, it identifies clusters where products have high overlap in features. Each resulting cluster can be summarized by its most common attributes, providing interpretable segments of products that reflect real-world style groupings.

Together, these methods provide a comprehensive analysis: Apriori captures local co-occurrence patterns between attributes, while dimensionality reduction and clustering provide global structure and segmentation.

Implementation Quality

The analysis was implemented entirely in Python using standard, open-source libraries suitable for categorical data analysis, clustering, and visualization. Key libraries included pandas for data handling, apyori and mlxtend for association rule mining, prince for Multiple Correspondence Analysis (MCA), scikit-learn for K-Means clustering and silhouette evaluation, and matplotlib for plotting results.

The implementation followed several core stages: We first applied the Apriori algorithm to extract frequent co-occurring attribute sets for products. Using the transaction representation where each product is a set of categorical attributes (season, gender, article type, color), the apyori library generated candidate rules.

We analyzed:

- Self-referencing rules where the antecedent overlaps with the consequent.
- Rules that do not include empty antecedents, with antecedents different from consequents.

Top rules were printed for inspection, displaying the antecedents and consequents. Additionally, seasonal subsets of transactions (summer, winter, spring, fall) were analyzed, comparing Jaccard's similarity of fashion trends across seasons and producing frequency tables of top itemsets per season. This approach allowed us to capture season-specific co-occurrence patterns instead of only looking at a broad overview.

To explore the global structure of the dataset, Multiple Correspondence Analysis (MCA) was applied to a dataframe with the categorical attributes. Using the prince library, we projected products into a 2-dimensional space, capturing the variance across all categorical attributes. The MCA transform was fitted to the dataset and the resulting coordinates were extracted for subsequent visualization and analysis of products. Coordinates of seasons, genders, and a few key colors or article types were plotted for a granular deep-dive, and a sample of coordinates were colored by season for a broader look at the underlying data.

K-Means clustering was applied to binary attribute data with reduced (10) dimensions with PCA. The resulting data captured a modest 59.2% of the global variance, helping identify groups of products with similar attribute profiles. Each cluster was summarized by counting and ranking the top values for gender, season, article type, and base color. This provided value counts and percentages of dominant attributes, revealing natural groupings in the dataset. The cluster summaries were printed for manual inspection.

The entire workflow is fully reproducible: Random seeds were fixed in MCA and K-Means to ensure consistent outputs. All steps, from transaction encoding, rule extraction, MCA projection, clustering, to visualization, are deterministic given the same input dataset. Modular functions in our notebook (`run_apriori`, `count_pct`) make it straightforward to apply the same pipeline to seasonal subsets or different categorical datasets. This implementation ensures that the analysis is transparent, systematic, and directly reproducible for further research or application to similar fashion dataset.

Rationale and Justification

We retained the three-part analytical framework, which consisted of association rule mining, dimensionality reduction, and clustering, but adopted more specialized and theoretically appropriate techniques within each stage. Instead of only running Apriori for the entire dataset, as implied in the proposal, running the algorithm separately on summer, winter, spring, and fall subsets also improved interpretability by avoiding the dataset's uneven year distribution and by isolating trends that are stable within a given season.

For dimensionality reduction, we strengthened the original plan by selecting MCA, which is designed specifically for multi-level categorical data. Clustering in our final workflow was implemented using

K-Means applied to a PCA-reduced binary feature matrix rather than MCA. PCA allowed us to compress thousands of binary features into a small set of components while preserving the dominant structure of attribute variation across products, while also providing enough dimensions for K-Means to form meaningful separations within the data. Therefore, we have expanded upon our original proposal by exploring not one, but two methods for dimensionality reduction for categorical values, with MCA having lesser dimensions of the two (2 vs 10 dimensions).

For clustering, we chose against analyzing many clear segments ($k=80-90$) for analyzing broader, but intentional and interpretable segments ($k=8$) (See Section 5: Results, Evaluation, and Visualization - *Evaluation and Validation*).

Overall, the final methodological choices improve upon the initial proposal. They provide more insights into how fashion attributes co-occur and organize, ultimately offering a more rigorous and interpretable analysis of the dataset.

5. Results, Evaluation & Visualization

Findings and Insights

Our overall results reveal strong patterns in how fashion products cluster based on categorical attributes. **Season emerges as the primary factor structuring the catalog.** Summer and fall items share substantial overlap, reflecting the extended warm-weather period in India, which encourages light clothing, t-shirts, and casual staples to remain relevant across both seasons. Winter and spring are more distinct, likely because colder temperatures drive the inclusion of accessories like watches, handbags, and layered outerwear in winter, while spring emphasizes lighter, style-focused items such as women's cosmetics and bright colors that align with romantic celebrations (Valentine's Day, Marriage occasions, etc) occurring around that time.

Gender also plays a major role in organizing products: men's and women's items form separate clusters, reflecting the inherently gendered nature of apparel, while unisex items, such as backpacks, occupy intermediate positions. Article types and colors reveal interesting patterns: neutral shades dominate staple clothing like men's t-shirts and shirts in summer and fall, while spring collections feature bright shades like pink, red, and purple, mostly in cosmetics. Cultural and stylistic trends further influence product composition; for example, women's kurtas appear in fall collections, likely linked to Indian festivals such as Diwali and Navratri.

Overall, the dataset demonstrates that fashion products naturally form coherent clusters across season, gender, category, and color. These high-level groupings directly address our primary research goal of uncovering which attribute combinations frequently occur together and understanding how fashion trends organize over time. By analyzing itemsets and identifying clusters (See Section 5: Results, Evaluation, and Visualization - *Visualization and Presentation* to see a method-wise analysis), we can see not only which items are commonly paired, but also how these patterns shift across the year, providing a structured view of co-occurrence in fashion data.

Evaluation and Validation

To validate the results, we combined quantitative metrics with qualitative inspection. We evaluated the Apriori component by tuning it with a fixed minimum support = 0.01, confidence = 0.2, lift = 1.0, maximum itemset length = 3, to ensure that only meaningful co-occurrences were retained.

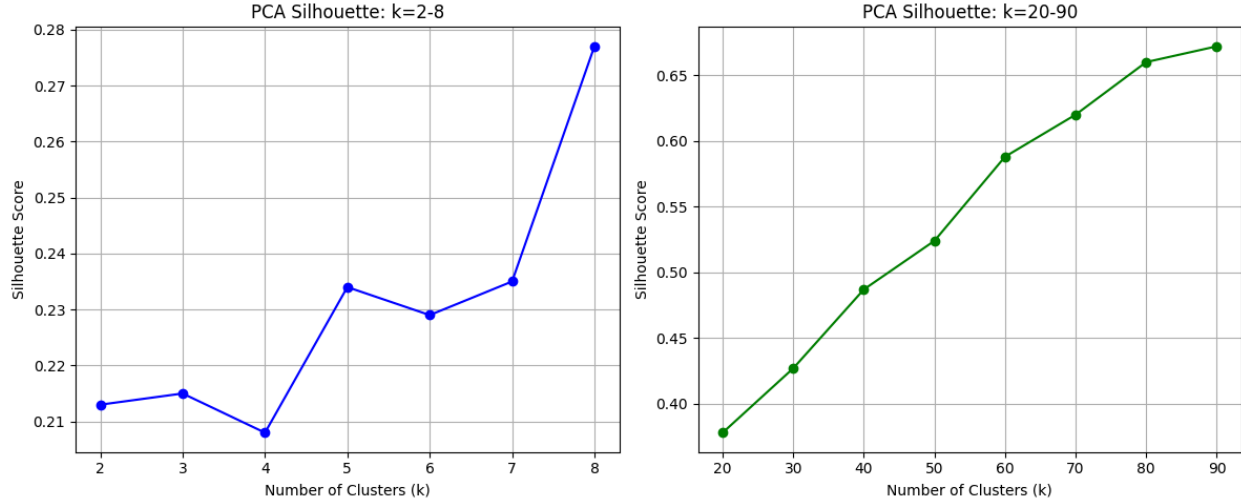


Fig 1: Two Silhouette score plots: 1.1) $k=2-8$ 1.2) $k=20-90$

Cluster quality was assessed using *silhouette* scores, which measured how closely products within a cluster resembled each other compared to other clusters. For K-Means clustering, we evaluated multiple values of k by computing silhouette scores, both for small k -values (2–8) and larger ranges (20–90), to determine the optimal number of clusters. While the clusters with the highest silhouette scores lay in the 20 - 90 range, we chose a k -value in the higher end of the smaller range: 8, with a weak silhouette score of 0.277. This was done so that the clusters, even if ambiguous, can be explained into different market trends.

Additionally, top itemsets, association rules, cluster summaries, and MCA plots were manually reviewed to verify that observed patterns reflected expected trends, including seasonal color palettes, gendered product types, and culturally relevant clothing styles.

Visualization and Presentation

The most frequent itemsets (Table 1) show a clear dominance of men's and summer products in the dataset. Nearly half of all items are tagged as men's (~50%) or summer (~48%), and women's items also appear often (~42%). Men's t-shirts show up in about 12% of products, and the combination t-shirts + summer appears in roughly 10%, highlighting how central warm-weather apparel is in this catalog. Colors like black are common (~22%), especially paired with men's items (~13%). Although winter items are less frequent overall, categories like women + winter still appear in about 10% of entries. Overall, these supports suggest that the dataset is heavily centered on men's and summer-oriented fashion.

Items	Support
(men,)	0.498537
(summer,)	0.483342
(women,)	0.419390
(fall,)	0.257316
(men, summer)	0.244778
(black,)	0.218981
(men, fall)	0.159216
(black, men)	0.132383
(men, tshirts)	0.118022
(women, winter)	0.103975
(tshirts, summer)	0.098708
(men, white)	0.069827
(tshirts, men, summer)	0.069219
(men, shirts)	0.064019
(white, summer)	0.063907

Table 1: Top 15 frequent itemsets (overall)

The top association rules from overall itemset analysis (Table 2) reveal clear and quantifiable co-purchase patterns across product categories, genders, and seasonal attributes. The strongest relationship appears between backpacks and unisex, where confidence reaches 87% and lift is an extremely high 17.9, showing that unisex items are far more likely to be backpacks than expected. Similarly, perfume and body mist with spring shows perfect confidence (100%) and a lift of 14.9, indicating a strong seasonal alignment of body care with springtime.

There are several winter-related patterns as well. Watches with winter have confidence values around 0.97 and lift near 5.1, while combinations like watches with men and winter or watches with black/white winter attributes show lifts ranging from 6.5 to 11.1. We can observe gendered fashion trends as well, such as heels predicting women winter with 77% confidence and lift of 7.4.

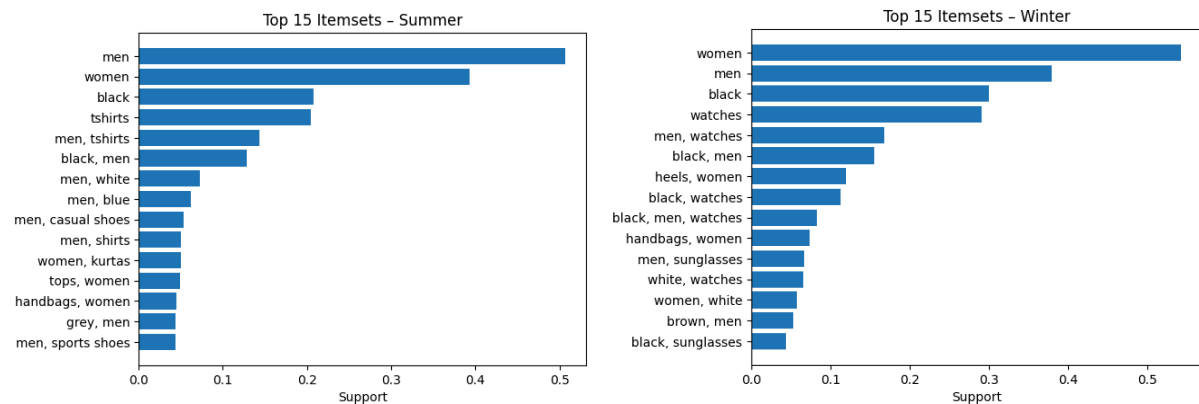
Antecedent	Consequent	Support	Confidence	Lift
(backpacks,)	(unisex,)	0.014204	0.871547	17.916521
(unisex,)	(backpacks,)	0.014204	0.291994	17.916521
(perfume,)	(spring,)	0.013799	1.000000	14.892390
(spring,)	(perfume,)	0.013799	0.205498	14.892390
(white, winter)	(watches,)	0.012628	0.633183	11.065505
(watches,)	(white, winter)	0.012628	0.220692	11.065505
(watches,)	(men, winter)	0.032145	0.561762	7.733416
(men, winter)	(watches,)	0.032145	0.442516	7.733416
(heels,)	(women, winter)	0.022983	0.771731	7.422250
(women, winter)	(heels,)	0.022983	0.221044	7.422250
(sunglasses,)	(men, winter)	0.012898	0.534017	7.351460
(jeans,)	(blue,)	0.010040	0.732348	6.615257
(watches,)	(black, winter)	0.021587	0.377262	6.559486

Antecedent	Consequent	Support	Confidence	Lift
(black, winter)	(watches,)	0.021587	0.375342	6.559486
(boys,)	(summer, tshirts)	0.010265	0.549398	5.565892
(women, watches)	(winter,)	0.019832	0.976718	5.094486
(watches,)	(winter,)	0.055691	0.973249	5.076392
(winter,)	(watches,)	0.055691	0.290478	5.076392
(black, watches)	(winter,)	0.021587	0.971631	5.067952
(men, watches)	(winter,)	0.032145	0.969450	5.056575

Table 2: Top 20 association rules (overall)

Looking at the seasonal itemsets individually (Fig. 2) gives a clearer picture of how shopping priorities shift throughout the year. In summer, the consistently high support for simple combinations like men + white or men + blue shows that neutral, easy-match colors are strong staples during warm months. In the winter, accessorizing is given more importance, as women’s handbags and men’s sunglasses enter the top itemsets.

Spring stands out as the only season where women’s color-related cosmetics repeatedly appear in the top itemsets, as pink, purple, and red shades, lipstick, and nail polish all show up with relatively high support, suggesting that makeup products peak around this time. Meanwhile, fall is the only season where women’s kurtas enter the top itemsets, pointing to a stylistic shift toward traditional clothing during transitional weather, possibly due to cultural Indian festivals like Diwali and Navarathri occurring around this time.



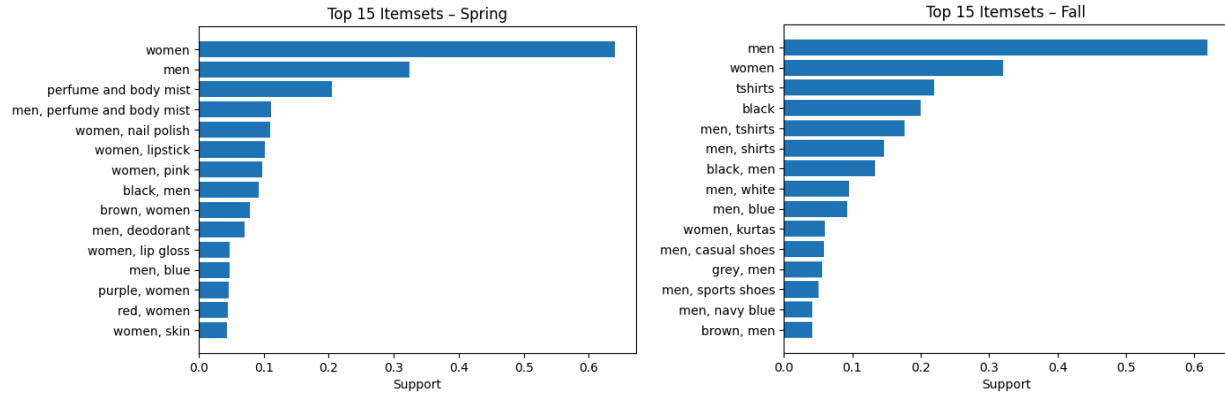


Fig 2: Top 15 frequent itemsets (season-wise)

Jaccard scores (Table 3) between seasonal itemsets show that Summer and Fall overlap the most (0.724), meaning many items work across both seasons. This aligns with India’s climate: Both summer and fall are warm because the country’s climate stays hot well past the monsoon season, with high temperatures extending from late spring through fall across most regions. Winter has minimal overlap with any other season (~0.16–0.22), reflecting its distinct, cold-weather product range. Spring shows the least overlap overall (~0.13–0.16), suggesting its items form a more unique, narrowly defined group.

Season	Summer	Winter	Spring	Fall
Summer	1.000	0.220	0.163	0.724
Winter	0.220	1.000	0.136	0.163
Spring	0.163	0.136	1.000	0.136
Fall	0.724	0.163	0.136	1.000

Table 3: Jaccard similarity between seasonal itemsets

The MCA plot (Fig. 3) provides a clear visual summary of how product attributes co-occur across the dataset. The most noticeable pattern is the strong seasonal separation: winter and spring fall far from summer and fall, the latter clustering tightly together. This mirrors the association-rule and Jaccard’s similarity results that warm-season items share many overlapping characteristics, while winter and spring have more distinct product profiles.

Gender attributes also organize meaningfully, with men and women forming somewhat separate regions surrounded by their typical items (e.g., shirts and blue for men, heels and pink for women). Unisex appears between the two, near neutral accessories like backpacks, matching its behavior in the rules. Colors such as black, white, pink and blue group near the summer - fall cluster, indicating shared styling or product trends there. Spring’s cluster is the most distinct, with springtime products (lipstick, and perfume and body mist).

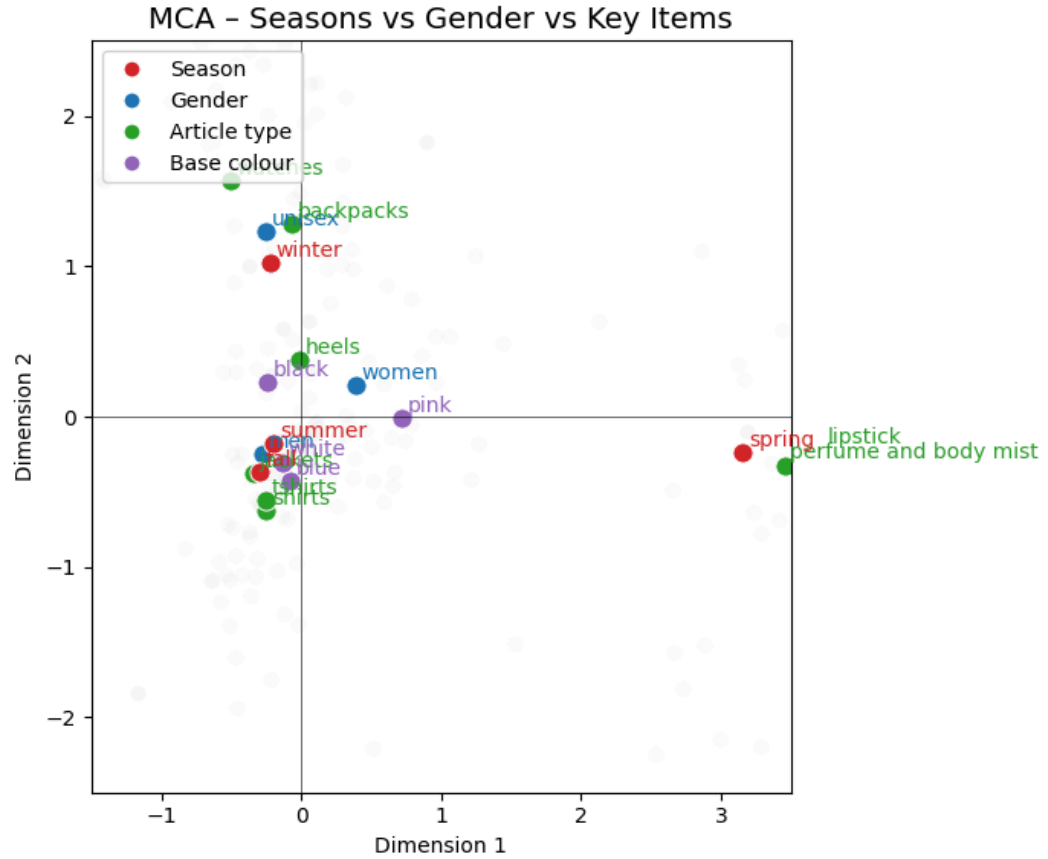


Fig 3: MCA plot of seasons, genders, and a few key article types

The K-Means clustering results primarily quantify and formalize patterns already observed in the dataset. Season emerges as the dominant factor, gender naturally separates products, certain article types like t-shirts, footwear, and accessories tend to co-occur, and colors align with expected seasonal palettes. What clustering adds is a more structured, segment-level perspective, grouping products into discrete clusters rather than just showing pairwise associations or visual proximities. It does not reveal fundamentally new trends beyond those already highlighted by association rules, itemsets, or MCA analyses.

Cluster	Top Gender (%)	Top Season (%)	Top Article (%)	Top Colour (%)
0	men (95.82%)	winter (94.5%)	watches (42.84%)	black (47.62%)
1	women (88.13%)	summer (63.73%)	handbags (11.76%)	black (15.98%)
2	men (85.34%)	summer (96.4%)	tshirts (96.02%)	white (15.65%)
3	men (97.13%)	summer (93.95%)	casual shoes (15.4%)	black (28.04%)
4	men (96.59%)	fall (99.45%)	tshirts (49.24%)	blue (18.65%)
5	women (89.79%)	winter (100%)	heels (19.94%)	black (21.15%)
6	men (75.97%)	fall (94.93%)	casual shoes (24.63%)	black (32.38%)
7	women (94.24%)	summer (66.72%)	kurtas (30.46%)	white (13.81%)

Table 4: Summary of eight product clusters obtained from K-Means clustering

6. LLM Use and Reflection

This project used OpenAI's ChatGPT models (GPT-5 and GPT-5.1) to assist with formatting and grammatical tasks. The prompt we used for reflection is as follows:

You are an expert data mining researcher. Please go through the report attached, and suggest three points of improvement for our proposal: The Hidden Palette: Mining Co-Occurring Fashion Attributes

The LLM suggested that the proposal could be strengthened in three main ways. First, it recommended clarifying the project's conceptual contribution by explaining what deeper insight is gained from uncovering co-occurring fashion attributes, beyond simply listing patterns. Second, it advised expanding the methodological justification, including why specific techniques such as Apriori, MCA, and K-Means were chosen over alternatives, why a silhouette score of $k=8$ was chosen over other values of k , and how dataset imbalance might influence the results. Finally, it suggested more explicitly connecting the findings to real-world fashion applications to highlight the practical value of the discovered patterns.

We already addressed many of these suggested improvements within the project's design and analysis. We intentionally selected techniques such as Apriori, MCA, and K-Means because they align with the core content taught in class, ensuring the methodology remains appropriate. Our choice of $k = 8$ was deliberate, prioritizing interpretable and meaningful clusters over higher-scoring but incoherent segmentations. We also connected our findings to real-world catalog curation by highlighting clear seasonal effects, such as winter's emphasis on accessories like watches and sunglasses, fall's inclusion of kurtas tied to cultural festivals, and spring's surge in beauty and cosmetic products, demonstrating how the uncovered patterns naturally inform merchandising and trend planning.

However, there are some suggestions on which we could conduct further analysis on, like how do class imbalances (e.g., men/summer dominance) bias support and clustering outcomes? Our analysis mostly focused on the big picture and our goal was to highlight these class imbalances. Future work on similar datasets (especially with larger attributes) can be conducted to further find hidden patterns.

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7. Jupyter Notebook & Reproducibility

Workflow

We can confirm that your notebook includes all key steps such as data processing, feature selection, model building, and analysis. Our workflow has already been covered by Section 3: Dataset & Representation - *Data Processing and Preparation*, Section 4: Analytical Techniques & Implementation - *Implementation Quality*, and Section 5: Results, Evaluation, and Visualization - *Evaluation and Validation*.

Clarity and Organization

High-level Markdown headings separate major phases, such as Association Rule Mining, Multiple Correspondence Analysis, and K-Means Clustering within PCA, each with analysis, evaluations, and findings, so readers can easily follow the pipeline. Within each section, descriptive markdown blocks explain the purpose of the code and summarize findings, while in-line comments clarify specific implementation choices.

Displayed Outputs

In this report, all figures from the notebook have been displayed as is in Section 5: Results, Evaluation, and Visualization—*Evaluation and Validation* and *Visualization and presentation*, with the exception of the MCA plot titled “MCA - Product Cloud Colored by Season.” We omitted this as Figure 3 gives a high level overview of the composition, supporting the notion that category centroids reflect the product distributions. However, you can refer to this figure in our notebook. Tabular/Matrix outputs have been displayed as markdown tables.

8. Supplementary Sections

Proposal Modifications / Dataset Overlap Clarification

Our dataset and project ideas do not overlap with other courses’ projects or any existing projects online or in other resources. This statement can be accepted as a pledge declaring the uniqueness and originality of our project.

Formatting and Structure

Our report strictly follows the formatting guidelines (page count, font, citation style, figure/table limits), as we have adhered to the template provided by the instructional team.

Roles & Collaboration

Team Member	Contributions
Robert Collis	Led primary data processing, applied core data mining techniques, created visualizations, and prepared the final project slides.
Lena Choi	Wrote the initial proposal, provided notebook code explanation and analytical commentary, managed Git, and served as the main code owner.
Vaishnavi Venkataraghavan	Conducted evaluation and validation of metrics, hyperparameters, and results, and prepared the final project report.

Ethics & Integrity

All analyses were conducted on product-level attributes only, ensuring that no user data or private records were involved. We credited all external sources, including datasets, libraries, and related research, to maintain academic integrity and avoided the use of proprietary or unlicensed material. Code, methods, and results were produced transparently and reproducibly, and no part of the work involved data manipulation, misrepresentation, or plagiarism. LLMs were used responsibly as a tool rather than a replacement for the team's original ideas for implementation and analyses.

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Addendum

As stated above, this was the ChatGPT prompt that was entered: *You are an expert data mining researcher. Please go through the report attached, and suggest three points of improvement for our proposal: The Hidden Palette: Mining Co-Occurring Fashion Attributes*

The response is as follows:

Here are **three targeted, substantive improvements** for your proposal based on the attached report:

1. Clarify the conceptual contribution beyond descriptive pattern-finding

Right now, the proposal presents strong results (itemsets, MCA structure, clusters), but it does not fully articulate *what new understanding* your analysis contributes beyond “these attributes co-occur.” Strengthening the framing — e.g., identifying *why* these co-occurrences matter, what latent structures they reveal (seasonality vs gender vs category dominance), or how symbolic attribute mining complements image-based fashion analysis — would give the project a clearer conceptual payoff.

2. Deepen the discussion of methodological choices and their limitations

Your pipeline is technically solid, but the reasoning behind certain decisions could be more explicitly justified. Examples:

- Why Apriori over FP-Growth given scalability?
 - Why MCA for visualization but PCA for clustering instead of using MCA for both?
 - Why $K = 8$ when silhouette scores suggest better separation at larger k ?
 - How do class imbalances (e.g., men/summer dominance) bias support and clustering outcomes? Expanding on these tradeoffs would demonstrate stronger methodological awareness.
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3. Strengthen the link from results to actionable or domain-specific insights

The proposal identifies clear patterns (e.g., warm-season similarity, winter accessorization, spring cosmetics), but it could benefit from explicitly translating these findings into *fashion-relevant implications*. For instance:

- How might retailers adjust seasonal catalog composition?
- How could designers use co-occurrence patterns to build palettes or curate collections?
- How could rule-based attribute structure support recommender systems or tagging pipelines?
Drawing these connections would make the project's impact clearer and elevate it beyond analysis into application.

If you want, I can also produce a polished paragraph-style version, a more critical version for peer review, or a shorter “reviewer-style” set of improvements.